

MVD Assignment Professional Summary

AutoML with PyCaret

Prepared for submission review

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Scope rule

This document summarizes the MVD assignment only. Extra-source Notebook 4 topics are intentionally excluded except where they are needed to explain the MVD workflow boundary.

3

MVD levels covered

3

Core notebooks
summarized

100%

MVD checklist mapped

0

Extra topics included

1. Executive Summary

This report summarizes the completed MVD assignment for Learning Day 15: AutoML with PyCaret. It is focused on the required Beginner, Intermediate, and Expert tasks only.

Main outcome

The MVD assignment demonstrates a complete AutoML workflow: fast model benchmarking, model tuning, model blending, save/load deployment readiness, explainability with SHAP, and fairness evaluation with mitigation. The strongest predictive result was achieved in the Breast Cancer classification task, while the Expert task showed why fairness and human review are necessary even when AutoML is used.

MVD Level	Notebook	Dataset / Use Case	Main Result	Business Meaning
Beginner	Notebook 1	Breast Cancer Wisconsin	Logistic Regression reached holdout AUC 0.9940 and Accuracy 0.9737.	AutoML can quickly create a strong medical-style classification baseline, but false negatives still need careful review.
Intermediate	Notebook 2	Logistics delay-risk proxy	Tuned Logistic Regression reached holdout AUC 0.9406; blend reached 0.9399.	Tuning and blending support operational delay-risk scoring and deployment readiness.
Expert	Notebook 3	HR attrition proxy	Fairness mitigation reduced Demographic Parity gap from 0.3081 to 0.0003.	AutoML must be checked for fairness before being used in sensitive HR decision support.

Dataset disclosure

The Beginner task used the sklearn Breast Cancer dataset. For the Intermediate and Expert tasks, the required Kaggle datasets were not available locally, so realistic proxy datasets were used to demonstrate the required MVD methods honestly and reproducibly.

2. MVD Requirement Coverage Matrix

The table below maps every MVD requirement to the completed notebook work. This section is the assignment proof-of-coverage.

MVD Requirement	Implementation Evidence	Status
Beginner: Load Breast Cancer Wisconsin dataset	Notebook 1 used the sklearn Breast Cancer dataset and prepared it for PyCaret classification.	Covered
Beginner: Use PyCaret setup()	Notebook 1 initialized the PyCaret classification experiment with the target variable and reproducible setup.	Covered
Beginner: compare_models() and report best AUC	Logistic Regression ranked first with cross-validated AUC 0.9971.	Covered
Beginner: create_model() and evaluate_model()	The selected Logistic Regression model was created and evaluated on holdout data.	Covered
Beginner: Confusion matrix and AUC curve	Confusion matrix and AUC-based evaluation were produced and interpreted.	Covered
Intermediate: Logistics/Supply Chain classification	A logistics-style delay-risk proxy dataset with 3,000 records was used because the Kaggle file was not available.	Covered with proxy
Intermediate: Top 3 models	Logistic Regression, LDA, and AdaBoost were selected as the top models by cross-validated AUC.	Covered
Intermediate: tune_model(n_iter=50)	Tuning was applied and performance before/after tuning was compared.	Covered
Intermediate: Blend top 3 models	A blended model was built and compared with tuned Logistic Regression.	Covered
Intermediate: Save/load model and REST API explanation	The final pipeline was saved, loaded, tested on new shipments, and linked to deployment logic.	Covered
Expert: IBM HR Attrition classification	An HR attrition proxy dataset with 2,500 records was used because the Kaggle file was not available.	Covered with proxy
Expert: SHAP / interpret_model	A Random Forest explainability model was created and used for interpretation workflow.	Covered
Expert: Fairness metrics by Gender	Demographic Parity and Equalized Odds were calculated for Gender groups.	Covered
Expert: Fairlearn mitigation	The notebook attempted official Fairlearn mitigation and used a compatibility-safe fallback when the local environment blocked ExponentiatedGradient.	Covered
Expert: Fairness-performance tradeoff	Baseline and mitigated models were compared using AUC, Accuracy, Demographic Parity, and Equalized Odds.	Covered

3. Beginner MVD Summary - Breast Cancer Classification

Notebook 1 demonstrates the basic PyCaret classification workflow: setup, model comparison, model creation, evaluation, and metric interpretation.

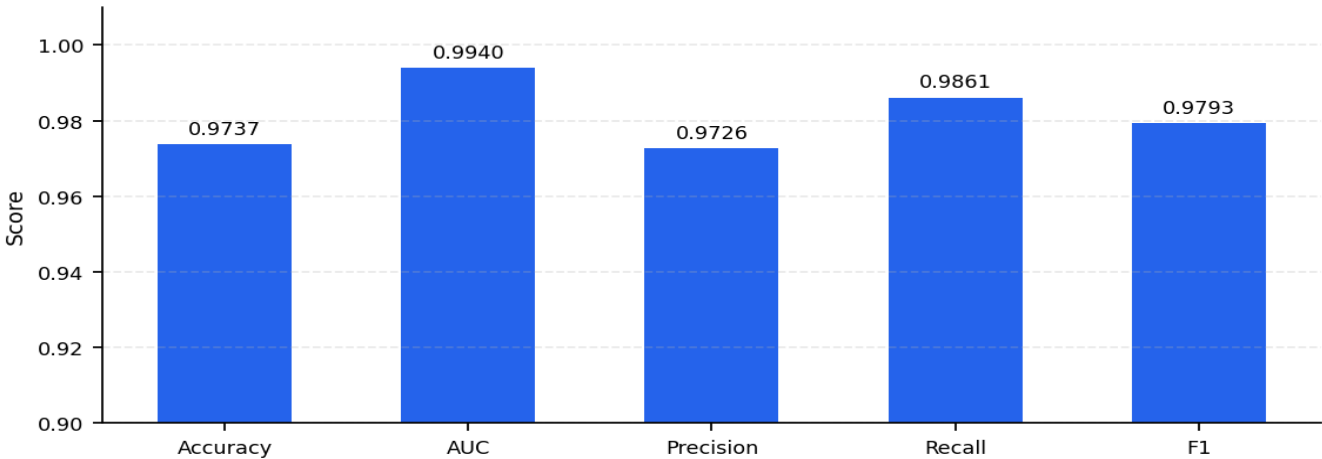
0.9971
Best CV AUC

0.9940
Holdout AUC

0.9737
Holdout accuracy

0.9861
Holdout recall

Notebook 1 - Holdout classification metrics



Item	Result
Best model	Logistic Regression
Cross-validated result	AUC 0.9971, Accuracy 0.9868, Recall 0.9965, Precision 0.9830, F1 0.9896
Holdout result	Accuracy 0.9737, AUC 0.9940, Precision 0.9726, Recall 0.9861, F1 0.9793
Confusion matrix summary	40 malignant correctly classified, 71 benign correctly classified, 2 malignant predicted as benign, 1 benign predicted as malignant
Business interpretation	The model is strong, but the two malignant-as-benign errors are important because false negatives carry the highest risk in a medical-style use case.

4. Intermediate MVD Summary - Tuning, Blending, and Deployment

Notebook 2 demonstrates model improvement and deployment readiness using a logistics-style delay-risk classification task.

3,000

Proxy records

0.9329

Best baseline CV AUC

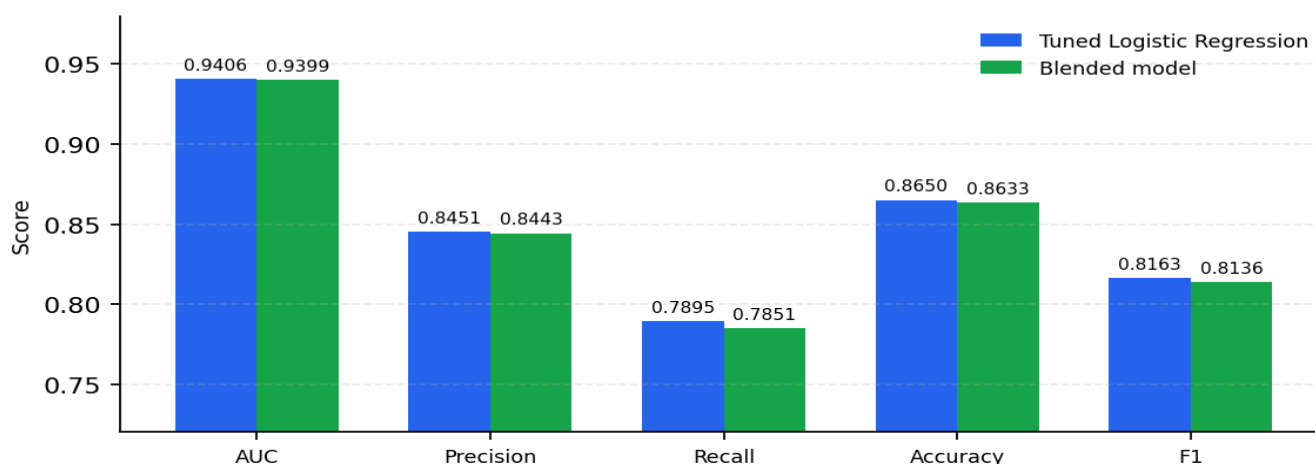
0.9406

Best holdout AUC

0.9399

Blend holdout AUC

Notebook 2 - Tuned LR vs blended model on holdout



Task	Completed Result
Top 3 models	Logistic Regression (AUC 0.9329), Linear Discriminant Analysis (AUC 0.9317), AdaBoost Classifier (AUC 0.9187)
Tuning	AdaBoost improved from AUC 0.9187 to 0.9269. Tuned Logistic Regression delivered the strongest final holdout AUC.
Blending	The blended model achieved holdout AUC 0.9399, very close to tuned Logistic Regression at 0.9406.
Save/load	The Logistic Regression PyCaret pipeline was saved and reloaded successfully.
New prediction test	New shipment predictions were generated; one shipment had a higher delay-risk score of 0.4258.
Business interpretation	The tuned model is the best single production candidate, while blending is useful as a robustness benchmark.

5. Deployment Readiness Summary

The Intermediate MVD required saving/loading and REST API understanding. This summary explains the practical deployment logic.

Deployment Step	What It Means	Why It Matters
save_model()	Stores the trained model and preprocessing pipeline.	The same preprocessing used during training is applied during production scoring.
load_model()	Reloads the saved pipeline later.	The model can be reused without retraining from scratch.
predict_model()	Scores holdout or new business records.	New shipments or customers can receive risk predictions.
REST API concept	A service receives input data, passes it through the saved pipeline, and returns predictions.	This makes the model usable by dashboards, web apps, or operational systems.
Monitoring	Track prediction quality, data drift, and business outcomes.	A deployed AutoML model must be checked over time, not treated as permanently correct.

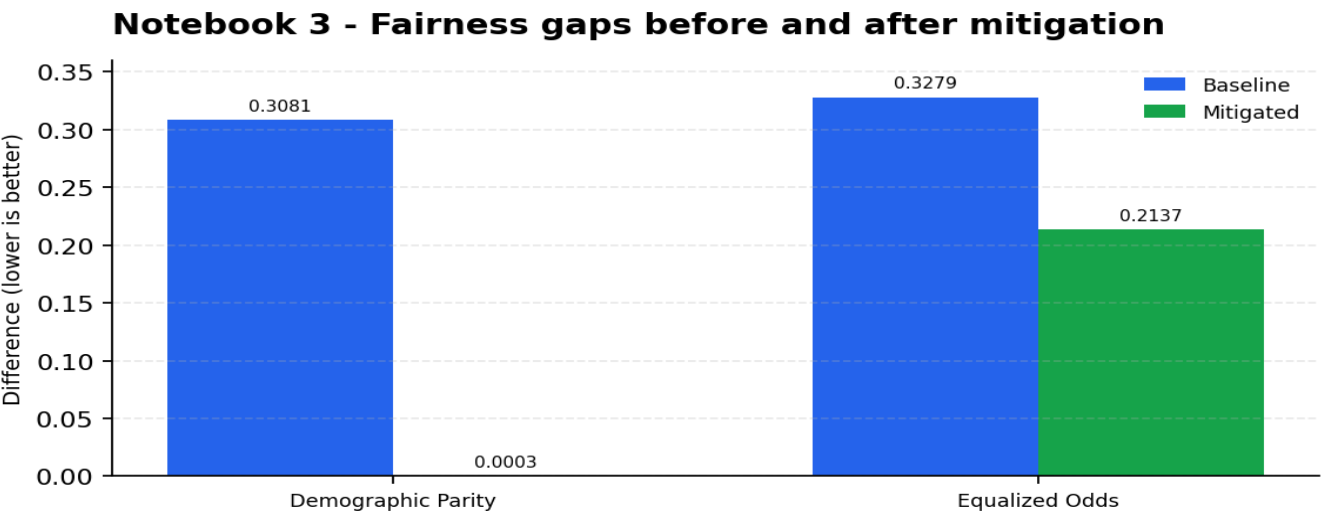
Deployment decision

For the MVD, the deployment target is not a full production platform. The assignment proves that the pipeline can be saved, loaded, and conceptually served through a REST API. A real production version would still need validation, versioning, monitoring, access control, and retraining rules.

6. Expert MVD Summary - SHAP, Fairness, and Mitigation

Notebook 3 moves beyond prediction quality and checks whether an HR attrition model behaves differently across Gender groups.

0.7091 Best PyCaret CV AUC	0.6808 Baseline LR AUC	0.3081 Baseline DP gap	0.0003 Mitigated DP gap
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Area	Result
Dataset	HR attrition proxy dataset with 2,500 rows; attrition rate was 4.4%, meaning the problem was strongly imbalanced.
PyCaret benchmark	Logistic Regression ranked best by AUC with cross-validated AUC 0.7091. Accuracy around 0.956 was misleading because the attrition class was rare.
SHAP / interpretation	A Random Forest explainability workflow was used to inspect feature influence and show why AutoML needs explanation before business use.
Baseline fairness	Female selection rate 0.5188 vs Male 0.2107; Demographic Parity Difference 0.3081 and Equalized Odds Difference 0.3279.
Mitigation	The local Fairlearn optimizer failed because of environment compatibility, so a compatibility-safe group-threshold fallback was used.
After mitigation	Demographic Parity improved from 0.3081 to 0.0003. Equalized Odds improved from 0.3279 to 0.2137. AUC stayed 0.6808 and Accuracy improved slightly from 0.6500 to 0.6560.

7. Expert Interpretation - Why This Matters

The Expert task shows that the MVD is not only about running AutoML commands. It is about responsible model review.

Concept	Meaning in This Assignment	Professional Interpretation
SHAP / interpretability	Used to understand which features influence attrition predictions.	Important because HR teams should not act on a black-box model without knowing the drivers.
Sensitive feature	Gender was used as the fairness grouping variable.	Predictions must be checked for group differences before being used in sensitive people decisions.
Demographic Parity	Measures whether groups receive positive predictions at similar rates.	The baseline gap was large. The mitigated model almost removed this gap.
Equalized Odds	Checks whether error behavior differs across groups.	The gap improved but remained imperfect, so the model would still need monitoring and review.
Fairness-performance tradeoff	The mitigated model changed fairness while preserving AUC.	The best model is not automatically the highest-AUC model; it must also be safe and explainable.

Responsible-use note

For an HR attrition use case, the model should support retention planning, workload review, and employee support. It should not be used as an automatic decision engine for hiring, firing, promotion, or punishment.

8. Final MVD Submission Checklist

This checklist confirms that the assignment requirements were completed and professionally interpreted.

- ✓ Beginner PyCaret setup completed on Breast Cancer classification.
- ✓ Best model identified using `compare_models()` and AUC.
- ✓ Selected model created, evaluated, and interpreted with holdout metrics.
- ✓ Confusion matrix and AUC-based evaluation interpreted with risk meaning.
- ✓ Intermediate logistics-style classification workflow completed with proxy data disclosure.
- ✓ Top 3 models selected and compared.
- ✓ Tuning applied and before/after performance discussed.
- ✓ Blended model created and compared against tuned model.
- ✓ Model save/load workflow demonstrated.
- ✓ REST API deployment logic explained at assignment level.
- ✓ Expert HR attrition workflow completed with proxy data disclosure.
- ✓ SHAP / interpretation workflow included.
- ✓ Fairness metrics calculated for Gender groups.
- ✓ Fairness mitigation applied with honest fallback explanation.
- ✓ Fairness-performance tradeoff explained using actual output values.

Final assignment conclusion

The MVD assignment is complete. The work demonstrates PyCaret AutoML from basic classification through tuning, blending, deployment readiness, model interpretation, and fairness-aware evaluation. The strongest predictive model came from Notebook 1, the most deployment-oriented workflow came from Notebook 2, and the most responsible-AI learning came from Notebook 3.

9. Key Metrics Appendix

This appendix provides a compact table of the most important numbers used in the professional summary.

Notebook	Metric	Value	Meaning
Notebook 1	Best CV AUC	0.9971	Best cross-validated model quality from <code>compare_models()</code> .
Notebook 1	Holdout AUC	0.9940	Strong unseen-data separation quality.
Notebook 1	Holdout Accuracy	0.9737	High overall classification correctness.
Notebook 1	Holdout Recall	0.9861	Strong detection of the positive class.
Notebook 2	Baseline LR CV AUC	0.9329	Best model from initial logistics benchmark.
Notebook 2	Tuned LR Holdout AUC	0.9406	Best final holdout result for logistics proxy.
Notebook 2	Blend Holdout AUC	0.9399	Very close to tuned LR, useful robustness check.
Notebook 2	Highest new-shipment risk score	0.4258	Operational scoring example.
Notebook 3	Best PyCaret CV AUC	0.7091	Best benchmark result on imbalanced HR proxy data.
Notebook 3	Baseline LR AUC	0.6808	Fairness baseline model performance.
Notebook 3	DP gap before / after	0.3081 -> 0.0003	Large improvement in selection-rate parity.
Notebook 3	EO gap before / after	0.3279 -> 0.2137	Improved but still not perfect error-rate fairness.

Excluded on purpose

Notebook 4 extra-source coverage is not part of this summary because the user requested MVD assignment only. Extra PyCaret modules, version comparison, and broader AutoML theory belong in the master theory document, not this MVD-only summary.